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# Cloud Security with AWS IAM

TS

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```
1 1
2 2 "Version": "2012-10-17",
3 3 "Statement": [
4 4 {
5 5   "Effect": "Allow",
6 6   "Action": "ec2:*",
7 7   "Resource": "*",
8 8   "Condition": {
9 9     "StringEquals": {
10 10      "ec2:resourceTag/Name": "development"
11 11     }
12 12   },
13 13 },
14 14 {
15 15   "Effect": "Allow",
16 16   "Action": "ec2:describe*",
17 17   "Resource": "*"
18 18 },
19 19 {
20 20   "Effect": "Deny",
21 21   "Action": [
22 22     "ec2:delete*",
23 23     "ec2:create*",
24 24   ],
25 25   "Resource": "*"
26 26 }
27 27 ]
28 28 }
```

+ Add new statement



**tshephomotlhale@outl...**

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## Introducing Today's Project!

In this project, I will demonstrate my learning journey in cloud security using AWS Identity and Access Management (IAM). I will start from scratch and gradually build the skills necessary to effectively work with this service.

### Tools and concepts

In this project, I learned AWS EC2 management, IAM users, groups, and policies, resource tagging, account aliases, and access control, ensuring secure, organized management of development and production environments.

### Project reflection

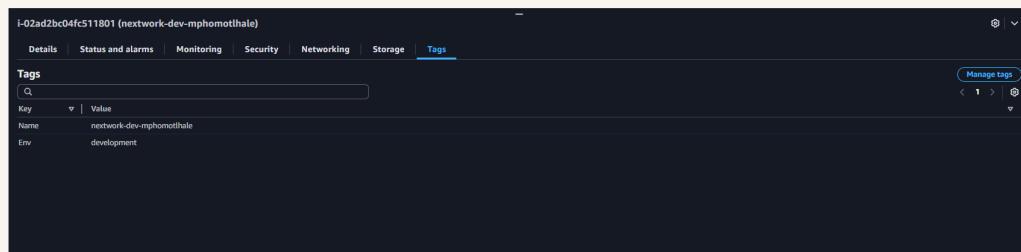
It took me approximately 2hrs to complete this project, including setting up EC2 instances, creating IAM users and groups, applying policies, tagging resources, and testing access controls.



# Tags

Tags in AWS act like custom labels that help you organize, identify, and manage your resources more efficiently. Each tag consists of a key (like “Name” or “Environment”) and a value (like “network-dev-mphomotlhale” or “development”) in my case

The tags I’ve used on my EC2 instances are Name and Environment. For the development instance: Name: network-dev-mphomotlhale Env: development For the production instance: Name: network-prod-motlhale Env: production





# IAM Policies

An IAM policy is a rule for who can do what with your AWS resources. It's all about giving permissions to IAM users, groups, or roles, saying what they can or can't do on certain resources, and when those rules kick in.

## The policy I set up

For this project, I've set up an IAM policy using JSON to define specific permissions for the development instance. This helps control who can access, manage, or modify the resource securely.

The effect of this IAM policy is to allow full EC2 access only to resources tagged with Env: development, while still allowing the user to view (Describe) all EC2 resources. It also explicitly denies permission to create/delete tags on any resource

When creating a JSON policy, you have to define its Effect, Action and Resource.

In a JSON policy, Effect defines whether access is allowed or denied, Action specifies the operations permitted, and Resource identifies the AWS resources those actions apply to.





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## My JSON Policy

```
Policy editor
1
2 "Version": "2012-10-17",
3 "Statement": [
4 {
5   "Effect": "Allow",
6   "Action": "ec2:*",
7   "Resource": "*",
8   "Condition": {
9     "StringEquals": {
10      "ec2:ResourceTag/Env": "development"
11    }
12  }
13 },
14 {
15   "Effect": "Allow",
16   "Action": "ec2:Describe*",
17   "Resource": "*"
18 },
19 {
20   "Effect": "Deny",
21   "Action": [
22     "ec2:DeleteTags",
23     "ec2:CreateTags"
24   ],
25   "Resource": "*"
26 }
27 ]
28 }
```

+ Add new statement



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## Account Alias

An AWS account alias is a custom, easy-to-remember name that replaces your 12-digit account ID, simplifying login URLs and making account access more user-friendly.

Creating an account alias took me a few minutes. Now, my new AWS console sign-in URL is <https://nextwork-alias-mphomotlhale.signin.aws.amazon.com/console>, making login easier and more memorable.

**AWS Account**

**Account ID**  
189886059746

**Account Alias**  
nextwork-alias-mphomotlhale [Edit](#) | [Delete](#)

**Sign-in URL for IAM users in this account**  
<https://nextwork-alias-mphomotlhale.signin.aws.amazon.com/console>

**Quick Links**

[My security credentials](#)

Manage your access keys, multi-factor authentication (MFA) and other credentials.



# IAM Users and User Groups

## Users

An IAM user group is a collection/folder of IAM users. It allows you to manage permissions for all the users in your group at the same time by attaching policies to the group rather than individual users.

## User Groups

An IAM user group is a collection/folder of IAM users. It allows you to manage permissions for all the users in your group at the same time by attaching policies to the group rather than individual users.

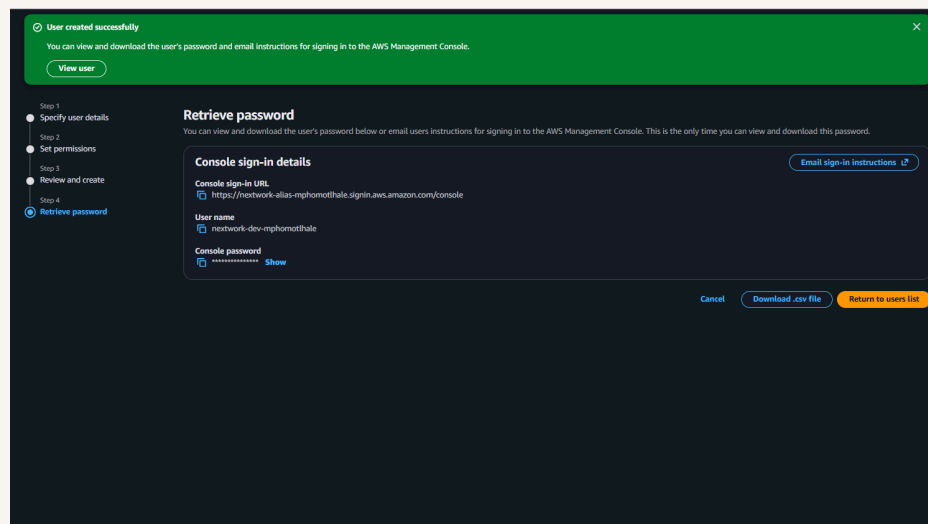
Attaching a policy to a user group grants all members the permissions defined in the policy, simplifying access management and ensuring consistent permissions across users.



## Logging in as an IAM User

You can share a new user's AWS sign-in details in two ways: Email: Send the user their username, temporary password, and console URL securely. Direct message: Provide the sign-in URL, username, and temporary password through password manager

The new IAM user's AWS dashboard shows access denied for most services, including Service Catalog, cost and usage info, forecasted costs, savings opportunities, and cost breakdowns, confirming permissions are correctly restricted.



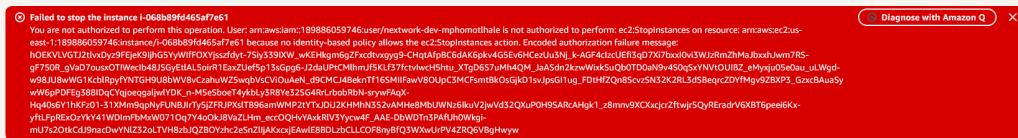


## Testing IAM Policies

I tested stopping and starting actions on my two EC2 instances to verify access: the development instance could be managed, while actions on the production instance were denied.

### Stopping the production instance

When I tried to stop the production instance, a banner appeared saying the action failed. This happened because the IAM user does not have permission to stop that EC2 instance, ensuring production resources remain protected.





# Testing IAM Policies

## Stopping the development instance

When I tried to stop the development instance, the action succeeded because the IAM user has the necessary permissions for resources tagged with Env: development.

A screenshot of the AWS Management Console. At the top, a green banner reads "Successfully initiated stopping of i-02ad2bc04fc511801". Below this, the "Instances (1/2)" page is shown. A filter "Instance state = running" is applied. The table lists two instances: "nextwork-dev-..." (ID: i-02ad2bc04fc511801) which is in a "Stopping" state, and "nextwork-pro-..." (ID: i-068b89fd465af7e61) which is in a "Running" state. The "nextwork-dev-..." instance is highlighted with a blue selection bar.

| Name             | Instance ID         | Instance state | Instance type | Status check      | Alarm status     | Availability Zone | Public IPv4 DNS         | Public IPv4 ... | Elastic IP |
|------------------|---------------------|----------------|---------------|-------------------|------------------|-------------------|-------------------------|-----------------|------------|
| nextwork-dev-... | i-02ad2bc04fc511801 | Stopping       | t3.micro      | 3/3 checks passed | User: arn:aws... | us-east-1c        | ec2-13-221-124-85.co... | 13.221.124.85   | -          |
| nextwork-pro-... | i-068b89fd465af7e61 | Running        | t3.micro      | 3/3 checks passed | User: arn:aws... | us-east-1c        | ec2-3-93-170-114.com... | 3.93.170.114    | -          |



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